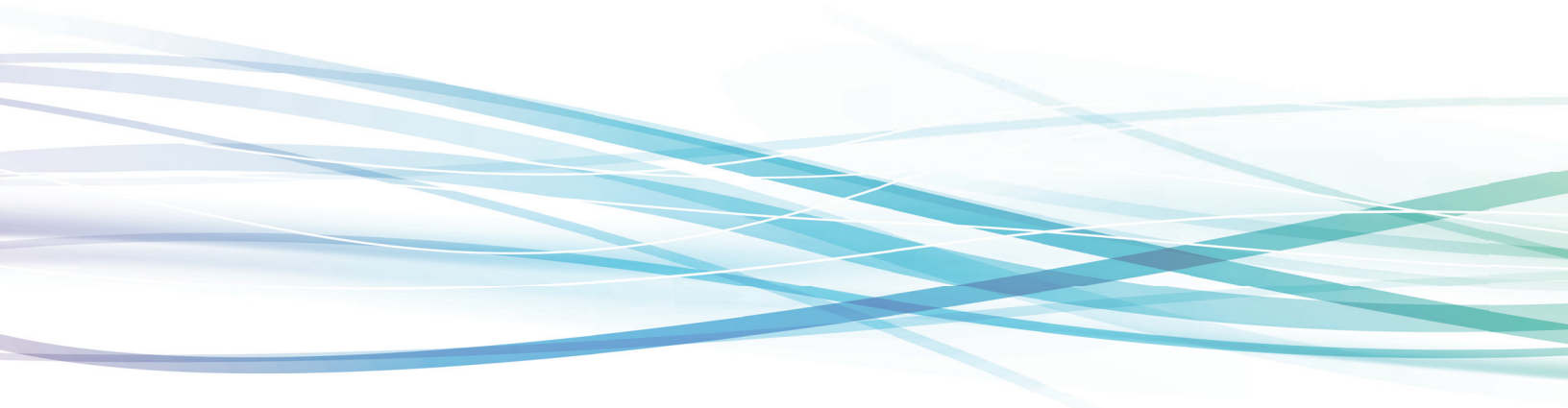


Polaris Tool Design Guide

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01	September 2020	Initial release of guide.
02	December 2022	Added information for tracking using markers on all faces and performed general updates.
03	May 2023	Defined Minimum Segment Length Constraint for Polaris Lyra and refactored information between User Guides and Tool Design Guide.

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Read Me First

This guide documents concepts and considerations for the design, construction, and characterization of tools that are used with NDI Polaris Systems.

Before designing a tool, you should understand:

- the information in this guide as it relates to your tool type
- your NDI system, as explained in the guide that accompanied it
- basic wiring concepts and techniques for electronic components, if you are developing active tools

Note **For help in understanding the concepts in this guide, or to discuss your tool design, contact NDI. See the contact information below.**

Warnings, Classifications, and Approvals

The user guide for your system includes a complete list of the warnings, classifications, and approvals that apply to your NDI system. Review this information before attempting to design a tool.

Disclaimers

NDI does not design tools, but provides this document to guide your design. Choices about material, ergonomics, design, application, tool functionality, and biocompatibility are your responsibility.

Some aspects of tool design are beyond the scope of this document and NDI's areas of expertise. This document cannot anticipate all concerns across the range of tools that can be developed.

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If you have any questions regarding the content of this guide or your NDI system, please contact us:



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Updates

NDI is committed to continuous improvements in the quality and versatility of its systems. To obtain the best results with your NDI system, and to ensure your design is current, check the NDI Customer Support Site regularly: https://support.ndigital.com.

1 Basic Concepts

This chapter introduces tools and markers. Before you design a tool for your Polaris system, review the user guide for your system.

1.1 Tool Overview

A tool is a rigid structure on which three or more markers are fixed so that there is no relative movement between them. A marker either emits or retro-reflects infrared light. Polaris optical measurement systems detect the infrared light and determine the 6D pose (position and orientation) of the tool within the measurement volume. The measurement system can then transmit the tool pose, marker positions, and status information to a host computer for collection, display, and further manipulation.

The three types of tools are described below.

1.2 Passive Tool Overview

Passive tools are wireless and incorporate retro-reflective passive markers. To track passive tools, the Position Sensor flashes infrared light from its illuminators. The light floods the surrounding area and reflects off the passive markers back to the Position Sensor.

There are three types of passive markers:

- **NDI Passive Spheres™** have a retro-reflective coating. The coating reflects infrared light back to its source instead of scattering it. NDI Passive Spheres snap-fit to the tool using NDI Passive Sphere mounting posts, which are manufactured to firmly hold NDI markers.
- **Radix™ Lenses** are hemispherical retro-reflective markers that can easily be wiped off if they become dirty. Before you can track a Radix Lens, you must incorporate it into a tool or mounting base (not provided by NDI). Radix Lenses are not compatible with NDI Passive Sphere mounting posts.
- **Reflective discs** are flat discs with a retro-reflective coating similar to NDI Passive Spheres. Because they are flat, there is some reduction in off-angle visibility.

1.3 Active Wireless Tool Overview

Active wireless tools are not electrically connected to the system. They incorporate active markers that are powered by battery or by other equipment they are attached to. To track active wireless tools, the Position Sensor pulses its illuminators in a way that is recognized by an infrared receiver on the active wireless tool. The active wireless tool detects the pulse and emits infrared light from the markers in response. The Position Sensor detects the infrared light the active markers emit.

1.4 Active Wired Tool Overview

Active wired tools are electrically connected to the System Control Unit by a cable and incorporate active markers. The System Control Unit controls the activation of the active markers. To track active wired tools, the Position Sensor detects the infrared light that the active markers emit. Active wired tools may also include visible light-emitting diodes (LEDs) and switches.

Only Polaris Systems connected to a System Control Unit can track active wired tools.

1.5 Comparing Tool Types

Table 1-1 Comparison of Tool Types

	Passive Tools	Active Wired Tools	Active Wireless Tools
Marker Type	NDI Passive Spheres Radix Lenses Reflective discs	Active markers	Active markers
Power Source	Not required	Wired connection to NDI system.	Battery or supporting equipment.
Design Complexity	Marker geometry must comply with unique geometry constraints.	Marker geometry can comply with unique geometry constraints or standard geometry constraints.	Marker geometry must comply with unique geometry constraints.
System Compatibility	All systems.	Systems connected to a System Control Unit.	All systems.
Electrical Safety		Requires wiring. May need to comply with the “Electrical Safety Considerations” on page 24.	Requires wiring. May need to comply with electrical safety considerations if the tool is patient-applied.

2 Tool Design Considerations

2.1 Determining the Right Tool for Your Application

Before you design a tool, consider the necessary and desired features of the tool. Table 2-1 provides a comparison of features.

To determine the right tool for your application, consider the following:

Tool Type Considerations

- Does the tool need to be wireless?
- Is the tool used for probing locations?
- Will the tool be used as a reference tool?
- Is the tool a calibrator or pivot block, used to determine the tool tip offset?

Design Considerations

- How much will the tool be rotated off-angle during use?
- Are markers needed on multiple sides of the tool?
- What is the required accuracy of the tool?
- What size does the tool need to be?
- Will this tool be used simultaneously with other tools?

Construction Considerations

- How will the tool be used (held by hand, attached to an object, etc.)?
- Will the tool require sterilization?
- Will the tool be reusable or disposable?

2.2 Marker Considerations

Table 2-1 provides a comparison of marker features.

Table 2-1 Marker Comparison

Marker Type	NDI Passive Sphere	Radix Lens	Active Marker
Disposable	✓	✓	
Wipeable		✓	✓

Table 2-1 Marker Comparison (Continued)

Marker Type	NDI Passive Sphere	Radix Lens	Active Marker
Sterilization	Available sterile and non-sterile	Designed for irradiation[1]	Designed for steam sterilization[1]
Cost	$	$$	$$$
Form Factor	Medium	Large	Small
Integration Effort	Medium	High	High
Maximum Marker Angle	90°	60°	60°
Reduced sensitivity to Partial Occlusion		✓	

[1] Radix Lenses and active markers are provided non-sterile. You must test and validate any method of sterilization you use. All packaging used to ship Radix Lenses and active markers is unsuitable for sterilization of any kind.

NDI does not offer a disc marker. The disposability, wipeability, cost, form factor, and integration effort vary depending on the materials and design used. The accuracy of disc markers generally deteriorates beyond marker angles of 45°.

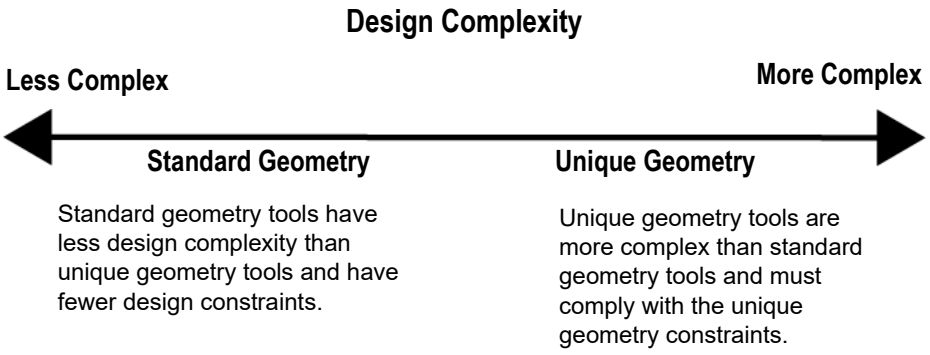
Marker Geometry

Marker geometry refers to how the markers are arranged on a tool. The geometry affects how the system identifies the tool.

- Standard geometry tools can have symmetric or non-unique arrangements of markers.
- Unique geometry tools require that markers have different segment lengths and angles between them so that the system can identify the markers and tools.
- If multiple unique geometry tools of the same tool type are used together, they must be compatible with one another.
- Standard geometry tools can be made smaller than unique geometry tools.

Active wired tools can use standard marker geometry or unique geometry. Passive and active wireless tools must use unique geometry.

When determining the best geometry for your application, consider your design complexity requirements and required acquisition time.



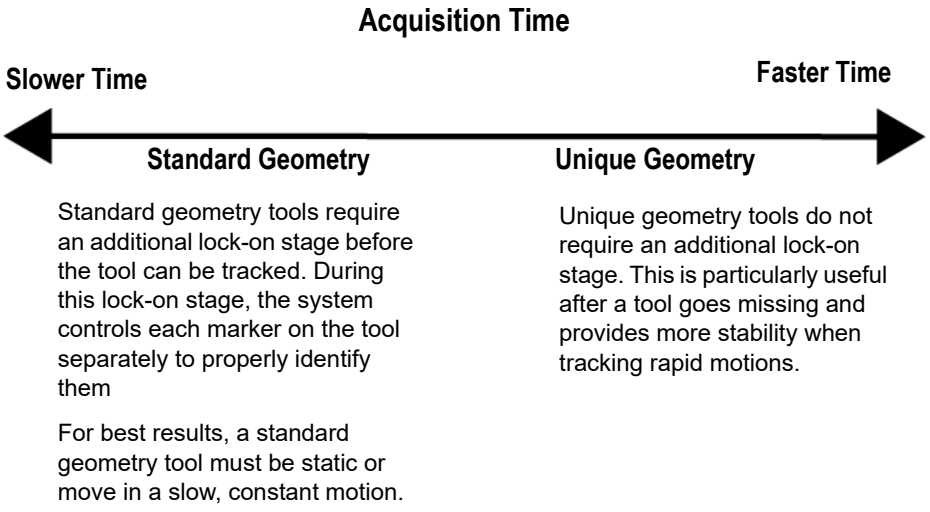
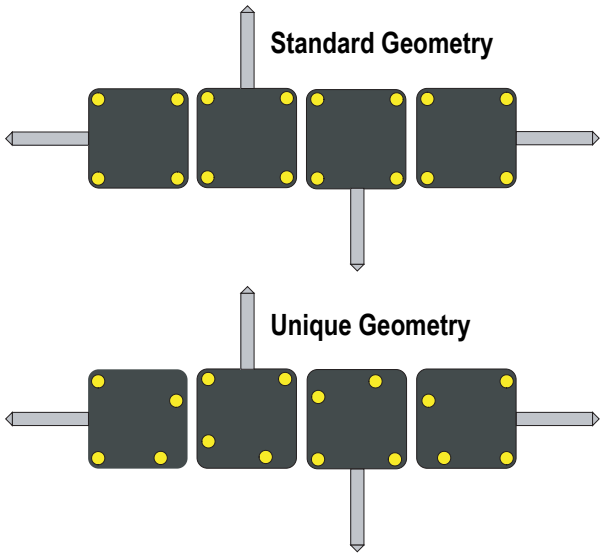


Figure 2-1 shows four orientations of a standard geometry tool and four orientations of a unique geometry tool. The Position Sensor must have previously identified each of the markers in order to identify the orientation of the standard geometry tool, but can determine the orientation of the unique geometry tool solely by marker geometry.

Figure 2-1 Standard and Unique Geometry Tools



2.3 Tool Design Best Practices

Consider the following best practices to design a tool that is as accurate as possible.

- A tool should have at least four markers. While the system can determine a tool transformation with three markers, a fourth marker adds redundancy. This additional information gives the Position Sensor more power when qualifying the transformation. For example, it is easier to detect partial marker occlusions with four markers than it is with three.
- Spread markers apart as much as possible. This reduces the impact of individual marker errors on tool orientation. Increasing the distance between markers can also reduce the chance of markers occluding each other. For the minimum segment lengths, see Table 4-1.
- Reference tools should be as large as possible. This reduces the effect of individual marker error on the tool transformation. Errors on reference tools are propagated to all tools that are tracked relative to this reference.
- The area of interest should be inside the reference tool, if possible. See “Points of Interest” on page 14.
- Markers along a probe axis should be furthest apart. This reduces the effect of individual marker error on the angular orientation of this axis.
- The tip on a probe should be as close as possible to the average position of the markers. See “Points of Interest” on page 14.
- The user should determine the tip offset of a probe before every use. This ensures that the tip is at the expected location relative to the markers and hasn't shifted due to handling or damage.

2.4 Tool Material

When choosing a material for the tool, consider the following material-related issues:

- Ergonomic factors such as weight, size, shape, ease of manipulation, tactility character.

- Rigidity of the material. Marker positions must remain fixed with respect to one another.
- The environment where the tool will be used. For instance, tools manufactured using ferrous material could cause interference if used in conjunction with MRI equipment.
- Sterilization and cleaning requirements of the tool. If the tool will require sterilization, consider what methods will be used. Certain sterilization methods can damage anodized metal.
- Reflectivity of the tool surface. A reflective tool can reflect infrared light from the illuminators back to the Position Sensor, which may interfere with tracking. This is usually not a concern for active tools.
- The backing on the marker. Markers should be backed by material that guarantees high contrast, so that the marker signal in the image is surrounded by a dark border.
- The opacity of the material. NDI recommends opaque material when possible. A translucent or transparent material on a multi-faced tool might allow the system to detect markers that are not facing the Position Sensor and are partially occluded, as described in “Marker Occlusion Issues” on page 41.

3 Design Concepts

3.1 Tool Definition File

Each tool needs a tool definition file to describe the tool to the system. The file is created with Cygna-6D (NDI-provided software) during the tool characterization process. For more information on that process, see “Characterizing Tools” on page 46.

A tool definition file uses the .rom extension. For passive and active wireless tools, you must provide this file to the system via the API. For active wired tools, you can use the API or the system can read the file from the tool’s serial read-only memory (SROM).

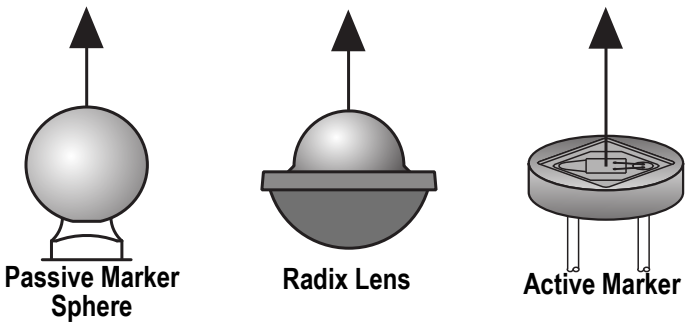
Tool definition files contain:

- manufacturer information, such as the manufacturer’s name and the tool’s part or serial number
- the number and type of markers
- the placement of markers relative to one another
- marker and face normal vectors
- tool tracking parameters
- definition of visible LEDs and switches, for active wired tools only

3.2 Marker Normal Vectors

A marker normal is a unit vector that defines the orientation of the marker with respect to the tool. Every marker on a tool has a marker normal. To improve the accuracy of the tool transformations, the system uses the marker orientation to restrict the viewability of certain marker types. For more information, see “Maximum Marker Angle” on page 33.

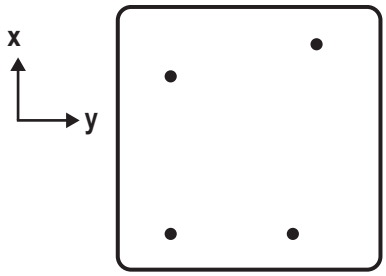
Figure 3-1 Markers with Marker Normal Vectors



Example: Setting Marker Normal Vectors for a Four-Marker Planar Tool

Consider the four-marker planar tool illustrated in Figure 3-2. The tool's local coordinate system is defined such that the markers lie in the *xy*-plane and point in the *-z* direction using a right-handed coordinate system.

Figure 3-2 Setting Marker Normal Vectors



Because the markers all point in the same direction (the *z* direction), the marker normal vectors for all four markers are the same and should be set to (*x*, *y*, *z*) = (0, 0, -1).

3.3 Faces

A face is a set of markers arranged on the surface of the tool such that the Position Sensor can view all markers assigned to the same face at the same time. Tools can have one to eight faces, and each marker can be assigned to a single face or to multiple faces.

Single-face tools are generally used when all markers are distributed on a plane with markers oriented in the same direction.

Multi-face tools are designed for greater rotational viewability; with these designs, markers are distributed on multiple surfaces of the tool, but only subsets of markers can be viewed at any given instant. When tracking a multi-face tool, the system can track faces using a single face only or using markers on all faces.

The multi-face tracking mode is selected on a tool-by-tool basis in the tool definition file. If a system that does not support "track all markers" has a tool definition file with this option enabled, the system will simply use the "track a single face" method.

Table 3-1 Multi-Face Tool Tracking Methods Supported by Polaris Systems

Polaris System	Track a Single Face	Track all Markers
Polaris Lyra®	✓	✓
Polaris Vega® (all models)	✓	Supported on CFR 009 and higher
Polaris Spectra®	✓	Not supported
Polaris Vicra®	✓	Not supported

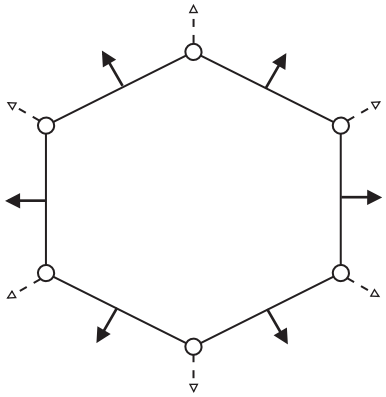
Track a Single Face of a Multi-Face Tool

In single-face tracking mode, the Position Sensor computes the reported 6D transformation using only markers assigned to the optimal face — the face most directly facing the Position Sensor. If the transformation calculated using the optimal face does not meet the tool tracking parameter criteria, the system will not track the tool.

The Position Sensor determines the optimal face based on the angles between the face normal vector and the sensors on the Position Sensor. To determine where the sensors are located on your system, see the user guide that accompanied your system.

A *face normal vector* is a unit vector that indicates the optimal direction that markers on this face can be viewed from. Face normal vectors are defined in the tool's local coordinate frame when the tool is characterized. Face normal vectors do not have to point in the same direction as the marker normal vectors that make up that face. For example, Figure 3-3 shows the marker normal vectors (dotted lines) and face normal vectors (solid lines) for a tool with six faces.

Figure 3-3 Face and Marker Normal Vectors



During tracking, if you rotate a tool so that a new face is the optimal face, the Position Sensor switches to use the markers from the new optimal face. This face switching occurs only when the angle between the new face and the Position Sensor is 2° less than the angle between the current face and the Position Sensor. This is known as *hysteresis* and is designed to prevent rapid face switching when multiple faces are at the same approximate angle to the Position Sensor.

When tracking active wired tools in this mode, only markers on the optimal face are activated. Therefore, segment lengths spanning multiple faces do not need to conform to the minimum segment length constraint.

Track all Markers on a Multi-Face Tool

NDI recommends using this tracking mode as it can improve the accuracy of tool tracking by increasing the number of markers used to compute the tool transformation. This tracking mode is also more robust if markers on the optimal face are occluded.

When tracking active wired tools in this mode, all markers on all faces are activated at the same time; therefore, segment lengths spanning multiple faces must conform to the minimum segment length constraint. Depending on the wiring matrix, this may also reduce the current and brightness of the active markers and can lead to increased noise in the 3D positions of the markers and can increase error in the reported 6D transformation. If this is undesirable, NDI recommends using the single-face tracking mode instead.

If you enable this tracking mode in the tool definition file, the Position Sensor uses all identified markers that meet the tracking criteria, regardless of which face the marker is assigned to.

For unique geometry tools, the system still uses face assignments to identify the tool. Therefore, at least three markers on a face must be visible at the same time to identify the tool. The unique geometry constraints apply, regardless of which multi-face tracking mode is used.

When tracking all markers on a multi-face tool, the system does not use the face normal vectors in the tool definition file. If you require backwards compatibility with Position Sensors running older firmware revisions, NDI recommends that you define face normal vectors for any multi-face tools that you intend to track using all the markers.

For active wired tools using standard geometry, there is no difference in functionality between tracking using all markers and simply assigning all markers to the same face.

3.4 Groups

Groups are sets of markers that are activated in separate frames. Groups are available only for active wired tools; active wireless and passive tools do not have control over individual markers and, therefore, do not support groups.

You assign each active marker to one of two groups during tool characterization. The groups are activated in subsequent frames and a 6D transformation is calculated using the marker positions from both groups only after the second group has been activated.

Because the groups are fired in subsequent frames, the firing sequence requires an extra frame. Depending on the setup, this may reduce the tracking rate. For example, if multiple active wired tools are tracked and all markers from these tools are assigned to the same group, the maximum tracking rate is the frame rate. However, if a new active wired tool is introduced that has markers assigned to the other group, or to both groups, the maximum tracking rate is ½ the frame rate. For more information on frame rate and tracking rate, see the user guide for your Polaris System.

Groups are primarily used to support the design of smaller tools. If you want to place markers closer together than the minimum segment length, by assigning the two markers forming that segment to either different faces or different groups, you can avoid the issue of markers merging.

Assigning markers to different faces to avoid markers merging works only when tracking using a single face on a multi-face tool. If you are tracking using all markers on the tool, then all faces are activated at the same time and the markers may still merge. However, if you have two markers that are closer than the minimum segment length, by assigning the markers to different groups, the system can use both markers to compute the 6D transformation.

Groups on a Single-Faced Tool

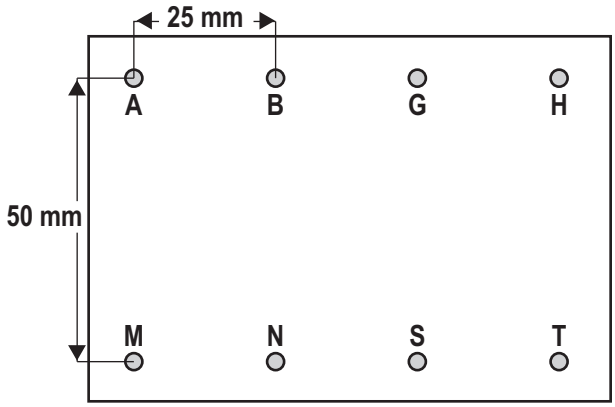
Figure 3-4 illustrates an 8-marker, single-faced tool. The marker letters correspond to the markers' positions in the tool wiring matrix on Figure 5-3.

The markers can be divided into faces and groups as follows:

- **Faces:** Since all eight markers are visible to the Position Sensor at the same time, this tool should be defined as a single face.
- **Groups:** The distance between columns of markers (25 mm) is less than the minimum segment length for all Polaris System measurement volumes. See "Minimum Segment

Length” on page 16. To avoid markers merging, adjacent columns of markers must be divided into two groups. One possible way to divide the tool into two groups is to put markers A, G, M, and S in group 1, and markers B, H, N and T in group 2. Notice that the markers in each group are evenly divided between the columns of the tool wiring matrix.

Figure 3-4 Groups on a Single-Faced Tool

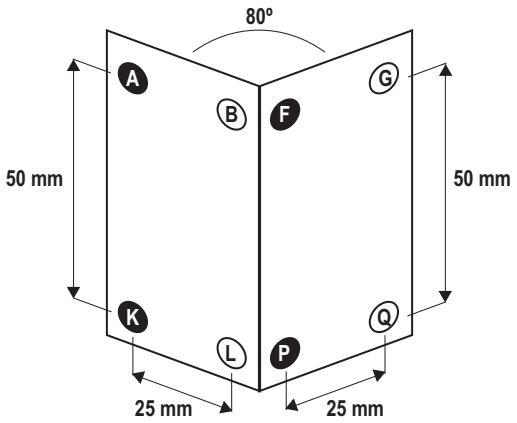


Groups on a Multi-Faced Tool

Figure 3-5 illustrates an 8-marker tool. The marker letters correspond to the markers’ positions in the tool wiring matrix on Figure 5-3. The markers can be divided into faces and groups as follows:

- **Faces:** The markers cannot all be seen by the Position Sensor at the same time, so this tool is divided into two faces. Markers A, B, K, and L belong to face 1; markers F, G, P, and Q belong to face 2.
- **Groups:** On each face, there are markers that are separated by less than the minimum segment length, so the markers must be divided into two groups. One possible way to divide the tool into two groups is to put markers A, F, K, and P in group 1, and markers B, G, L, and Q in group 2. Notice that the markers in each group are evenly divided between the columns of the tool wiring matrix.

Figure 3-5 Groups on a Multi-Faced Tool



3.5 Coordinate Systems

Global Coordinate System

The Position Sensor's global coordinate system is described in detail in the user guide that accompanied your system. The global coordinate system is defined during manufacturing and cannot be changed. The global coordinate system has an origin located at the Position Sensor and three axes (X, Y, Z). The system reports the transformations of tools in the global coordinate system. If two tools are tracked in the Measurement Volume, then one of the tools can be used as a reference; host software can calculate and report the relative transformation of the tracked tool in the local coordinate system of the reference tool.

Local Coordinate System

Each tool has a local coordinate system consisting of an origin and three axes (x, y, z). This right-handed coordinate system is defined during tool characterization using Cygna-6D. When the system reports the pose of a tool, the position and orientation describe the relative transformation of the tool local coordinate system relative to the Position Sensor's global coordinate system. The marker positions, marker normal vectors, and face normal vectors are defined in the tool definition file using the tool's local coordinate system.

Tips for defining the local coordinate system:

- Be consistent in defining local coordinate systems for a suite of tools; this makes application development easier. For example, it is conventional on a probe to align the z-axis to the probe shaft, with the positive z-axis pointing away from the tool tip.
- You can define a tool's origin at the tip of a probe however, if the tool is later bent, the origin is no longer located at the tip. NDI recommends pivoting the tool or using a calibrator to determine the tool tip offset before each use.

3.6 Linear Tools

A linear tool is created if all markers that are used to determine the 6D transformation are approximately collinear. When this happens, the rotational stability of the 6D transformation about the collinear axis is reduced and may lead to inaccurate transformations. Generally, the rotational stability about any arbitrary axis passing through the centre of mass of the tool increases the further the markers are from this axis.

Designing a linear tool may not be a problem if the tool only needs to track five degrees of freedom (5DOF) and the roll about the collinear axis is ignored. However, in this case note that the Position Sensor always returns a 6D transformation; if a rotational axis is to be ignored, the host application will need to handle this appropriately.

If a collinear tool is intentionally designed, do not assign marker or face normal vectors, as these vectors are not valid for 5DOF tools. Assigning marker normal vectors to a 5DOF tool's markers may cause intermittent tracking by erroneously reporting markers off-angle. Without marker normal vectors, the maximum marker angle cannot be used. See "Tool Tracking Parameters" on page 32. Therefore, collinear tools should not be designed with active markers or Radix Lenses as they should have a maximum marker angle of 60° defined in their associated tool definition file.

For tools where there are more markers than the minimum number of markers needed to track the tool, consider all subsets of markers that could be used to report a transformation when assessing

collinearity. The minimum number of markers and the spread parameters described in "tool tracking parameters" may be helpful for avoiding collinear markers.

3.7 Three-Marker Locking

Before the Position Sensor can track a tool, it must be able to identify the tool based on the visible markers. For unique geometry tools, it does this by comparing segment lengths and angles; for standard active wired tools, the System Control Unit activates individual markers sequentially. If a tool becomes occluded by an operator, another device, or another tool, the Position Sensor must re-identify the tool before it can report pose information. The system reports the tool pose only if all the tool tracking parameters are met.

By default, if a transformation is determined, but considered invalid because it does not meet the tool tracking parameters, then the system discards the tool's transformation. The Position Sensor must re-identify the tool before it can calculate the transformation again.

If the three-marker locking option is enabled during tool characterization, when a transformation is considered invalid, the system continues to internally track the tool. Because the system is still tracking the markers that are visible, it does not need to re-identify the tool when the tool tracking parameters are met.

NDI recommends enabling three-marker locking, unless three of the markers are collinear. In this case, the internally tracked transformation will be unstable about the collinear axis and the system may not properly identify the fourth marker when it becomes visible again.

Example

Consider a four-marker tool with the three-marker locking option enabled and where the system can detect only three of the markers on the tool.

- If the minimum number of markers is set to three, the system continues to track the tool and report transformations.
- If the minimum number of markers is set to four, the system continues to track the tool but reports the tool as MISSING. When the system detects the fourth marker, it calculates and reports the transformation without having to re-identify the tool.

3.8 Points of Interest

Points of interest are points on or near a tool where accuracy is most important. Points of interest are determined by the tool's intended use. When designing a tool, it is important to understand and consider the relationship between points of interest, the origin of the tool's local coordinate system, and the resulting transformations that the system will return. The aim is to design a tool that produces the greatest accuracy at the points of interest.

Points of interest are an important consideration when placing markers on the tool. The most accurately tracked point is the average position of all the individual markers used to calculate the transformation. The closer the average position of the markers to the points of interest, the greater the accuracy at the points of interest. Ideally, the origin of the local coordinate system, the average position of the markers, and point of interest are all the same. However, this is not always possible and in some cases not even advisable.

Consider a simple probe. You may not be able to place the markers around the probe's tip because of physical restrictions. However, placing the markers as close to the tip as possible will minimize the offset vector required and improve accuracy. When the point of interest lies far from the average position of the markers, it is beneficial to determine the offset vector from the origin of the local coordinate system to the point of interest each time the tool is used; this can be accomplished using a calibration fixture or by pivoting the tool about the point of interest.

The point of interest must be rigidly fixed relative to the markers.

3.9 Axes of Interest

Similar to a point of interest, an axis of interest represents an imaginary vector in (or projected from) a tool where measurement accuracy is most important. For example, if the trajectory of a probe must be in a particular direction before being used, the axis of interest is an imaginary line traveling along that trajectory.

The same criteria discussed for points of interest also apply to axes of interest. You must consider axes of interest when placing markers and defining the tool's local coordinate system. The closer one of the markers is aligned with an axis of interest, the greater the accuracy of that axis of interest.

4 Design Constraints

All tools must have:

- a tool definition file
- between 3 and 20 markers
- a maximum of 8 faces

This chapter describes the constraints related to the number and arrangement of markers on a tool to ensure that the Polaris System can identify and track the tool.

4.1 Minimum Segment Length

Your tool design must maintain a minimum distance between markers, as specified in Table 4-1.

Table 4-1 Minimum Segment Length Between Markers

Position Sensor	Measurement Volume	Minimum Distance
Polaris Lyra	Pyramid	30 mm
	Extended Pyramid	40 mm
Polaris Vega	Pyramid	40 mm
	Extended Pyramid	50 mm
Polaris Spectra	Pyramid	40 mm
	Extended Pyramid	50 mm
Polaris Vicra		30 mm

The minimum distance between markers allows the Position Sensor to distinguish the markers throughout the characterized measurement volume, at an angle of 45º at the back corner of the measurement volume. If the markers are closer together, the system may not track the tool reliably throughout the measurement volume because the marker images may merge, as described in “Marker Occlusion Issues” on page 41.

For passive and active wireless tools, the minimum segment length constraint applies to all markers on all faces of the tool.

For active wired tools, the minimum segment length constraint applies to both standard and unique geometry tools. However, markers that are activated at different times do not have to adhere to this constraint. You can accomplish this by assigning markers to different faces or by assigning markers to separate groups. If you assign markers to different faces so that they can be closer together than the minimum segment length, only a single face can be tracked at a time. For more information about tracking modes, see “Track a Single Face of a Multi-Face Tool” on page 10.

4.2 Unique Geometry

Minimum Segment Length Distance

For a unique geometry tool, the segment lengths of all pairs of markers on the same face must differ by at least 3.5 mm. This is because a unique geometry tool is identified based solely on the relative arrangement of the markers. You can also assign a maximum of six markers to the same face.

Passive and active wireless tools must have unique geometries. Active wired tools can be classified as either standard geometry or unique geometry.

The minimum segment length difference constraint does not apply to standard geometry active wired tools.

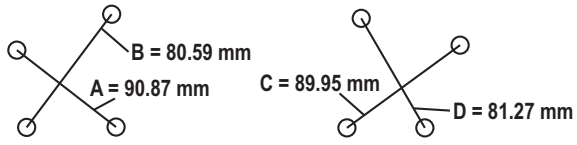
Tool and Face Compatibility

When multiple unique geometry tools are tracked at the same time or when a multi-face unique geometry tool is used, an additional constraint is required to ensure that the system can correctly identify individual tools and faces. The following terminology is necessary for understanding the tool and face compatibility constraint.

- *Segment length*: The distance between two markers on a tool
- *Segment pair*: Two segments on the same face
- *Segment angle*: The angle formed by a segment pair
- *Like segments*: Segment lengths that differ by less than 3.5 mm; due to the minimum segment length difference constraint, like segments exist on different faces or tools
- *Like segment pairs*: Two segment pairs, each on a different tool or face of the same tool type, and where each segment in one pair has a like segment in the other pair

Figure 4-1 illustrates like segment pairs. Segments A and B form one segment pair, and segments C and D form another segment pair. A and C are *like segments* and B and D are *like segments*, so segment pairs AB and CD are *like segment pairs*. The segment angles between AB and CD must differ by more than 2°.

Figure 4-1 Like Segment Pairs



For tools and faces to be compatible, like segment pairs must have segment angles that differ by at least 2°. You can determine the segment angle (θ in rad) between two segments *v1* and *v2* using the dot product:

$$\cos\theta = \frac{\mathrm{dot}(\vec{v1}, \vec{v2})}{|v1||v2|}$$

Since passive, active wireless, and active wired tools are tracked in separate frames, you can use the same tool geometry for each of the tool types.

To identify active wired tools using the unique geometry method, only a single face is activated at a time; therefore, active wired tools can reuse the same marker geometry on multiple faces and across multiple tools.

Mirror-image tools, where the marker geometries of two tools differ only in a reflection across an axis, are incompatible as they do not satisfy the tool compatibility constraint.

Marker Count and Unique Geometry

As the number of markers on a tool or face increases, the following aspects of tool design become more complicated:

- The number of segments within a face is determined by *n* choose 2, where *n* is the number of markers
- The size of the tool increases. All segments within a face must differ by 3.5 mm; therefore, the minimum length of the longest segment is determined as ($x + y^*(s\text{-}1)$), where *x* is the minimum segment length, *y* is the minimum segment length difference, and *s* is the number of segments.
- The number of segment pairs and angles are determined by *s* choose 2, where *s* is the number of segments.

Table 4-2 shows these effects:.

Table 4-2 Effects of Using More Markers

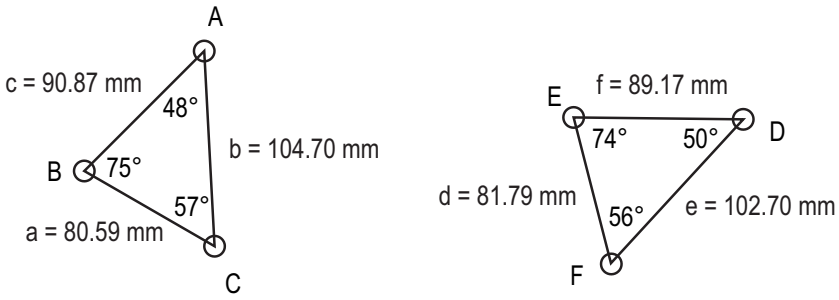
Number of Markers on a Face	Number of Segments	Minimum Length of Longest Segment	Number of Segment Pairs
3	3	($x + 2y$) mm	3
4	6	($x + 5y$) mm	15
5	10	($x + 9y$) mm	45
6	15	($x + 14y$) mm	105

Example: Checking Tool and Face Compatibility

Figure 4-2 shows two faces or tools, each with three markers (A, B, C and D, E, F). The three segment pairs *ab*, *bc*, and *ca* fail the compatibility check because they each form a similar segment pair with *de*, *ef*, and *fd* and the segment angles differ by less than 2°. Even if only one of the similar segment pairs failed, the tools or faces would be incompatible.

Figure 4-3 shows two similar faces or tools; however, these two faces or tools are compatible with each other. The similar pair *bc* and *ef* have segment angles which differ by more than 2°, and segment *a* does not form a similar segment amongst the markers D, E, and F.

Figure 4-2 Incompatible Faces or Tools

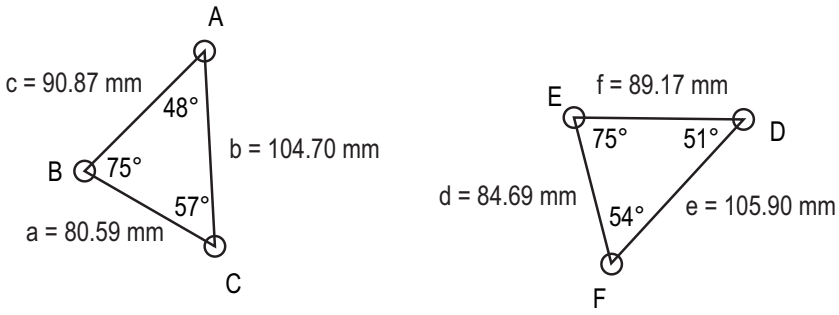


|a - d| < 3.5 mm
|b - e| < 3.5 mm
|C - F| < 2.0°

|b - e| < 3.5 mm
|c - f| < 3.5 mm
|A - D| < 2.0°

|c - f| < 3.5 mm
|a - d| < 3.5 mm
|B - E| < 2.0°

Figure 4-3 Compatible Faces or Tools



|a - d| > 3.5 mm
|b - e| < 3.5 mm
|C - F| > 2.0°

|b - e| < 3.5 mm
|c - f| < 3.5 mm
|A - D| > 2.0°

|c - f| < 3.5 mm
|a - d| > 3.5 mm
|B - E| < 2.0°

4.3 Summary of Design Constraints

Table 4-3 summarizes how the tool design constraints apply to different tool types:

Table 4-3 Tool Design Constraints

	Passive Tools	Active Wireless Tools	Active Wired Tools (w/ Unique Geometry)	Active Wired Tools (w/ Standard Geometry)
Tool Definition File	Required	Required	Required	Required
Maximum Number of Markers	20	20	20	20
Maximum Number of Faces[1]	8	8	8	8
Marker Geometry	Unique Geometry	Unique Geometry	Unique Geometry	Standard Geometry
Number of Markers per Face	3-6 NDI recommends at least 4	3-6 NDI recommends at least 4	3-6 NDI recommends at least 4	3-20 NDI recommends at least 4
Groups	Not Supported	Not Supported	Supported	Supported
Minimum Segment Length	Applies to all markers on a tool within and between faces	Applies to all markers on a tool within and between faces	Applies to all markers on a single face that have the same group[2]	Applies to all markers on a single face that have the same group[2]
Minimum Segment Length Difference	Applies to all segments within a face	Applies to all segments within a face	Applies to all segments within a face	Does not apply
Tool and Face Compatibility	Each segment pair within a face cannot be too similar in length and segment angle with any other markers on the same tool or on another passive tool	Each segment pair within a face cannot be too similar in length and segment angle with any other markers on the same tool or on another active wireless tool	Does not apply	Does not apply

[1] Each marker can be assigned to one or more faces.

[2] If you are tracking using all markers on a face, the minimum segment length applies to all markers within and between faces in the same group.

5 Markers and Tools

This chapter offers information about the markers and tools that you can use with NDI Polaris systems.

5.1 Optical Markers

NDI Passive Spheres

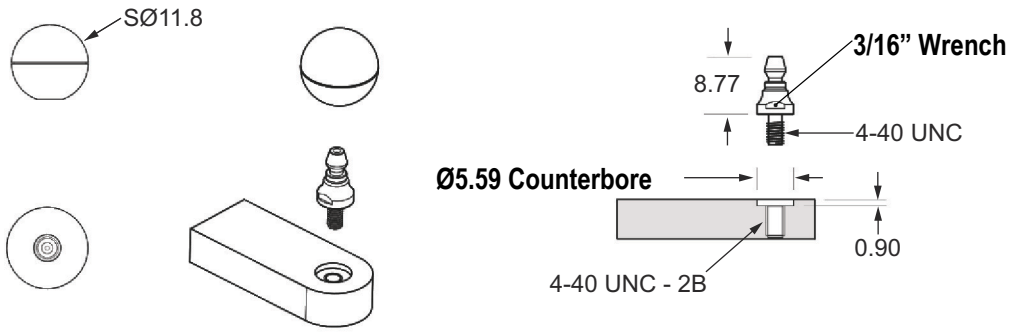
NDI Passive Spheres were designed to meet the needs of Polaris systems and are used to characterize every Polaris system. NDI Passive Spheres are the most common optical marker used with Polaris systems due to their wide viewing angles and ease of use.

When you snap an NDI Passive Sphere on to an NDI Passive Sphere mounting post, you will hear an audible snap indicating that the marker is properly engaged. When correctly fitted to the mounting post, the measured position of the marker is at the centre of the top of the mounting post. The mounting post was designed so that this position can be measured with a 1 mm diameter ball tip probe.

When mounting an NDI Passive Sphere:

- Use the base of the post as a datum in a pocket.
- Use the threaded part of the post to hold the post in place, not to control the height of the post.
- The base of the post includes wrench flats to help tighten the post into a pocket.

Figure 5-1 NDI Passive Sphere and Post



This drawing is provided as is, for reference purposes only. The drawing is not to scale. All dimensions are in mm. All dimensions are approximate.

NDI Passive Spheres are sterilized; however, non-sterile passive spheres are also available from NDI. For more information on package configurations, contact NDI. See “Contact Information” on page iii.

Radix Lenses

Radix Lenses are passive markers with two hemispherical lenses that form a retro-reflector. Radix Lenses are ideal for applications where markers may contact liquids or debris, such as in orthopedic

surgery. When contaminated, Radix Lenses are more accurate than other similarly contaminated optical marker types. If they become contaminated with liquids or particulate, you can wipe the Radix Lens clean to resume tracking.

The optical centre of the Radix Lens is the centre of the top hemisphere, 2 mm above the bottom surface of the flange. Unlike NDI Passive Spheres, Radix Lenses must be permanently integrated into an optical tool before you can use them. For details on designing optical tools with Radix Lenses, see the *Radix Lens Integration Guide*.

Radix Lenses are provided non-sterile. You must test and validate any method of sterilization you may wish to use.

Reflective Discs

Passive disc markers are made from circular cut-outs of flat retro-reflective materials. The measured position of a disc marker is the centre of retro-reflective material.

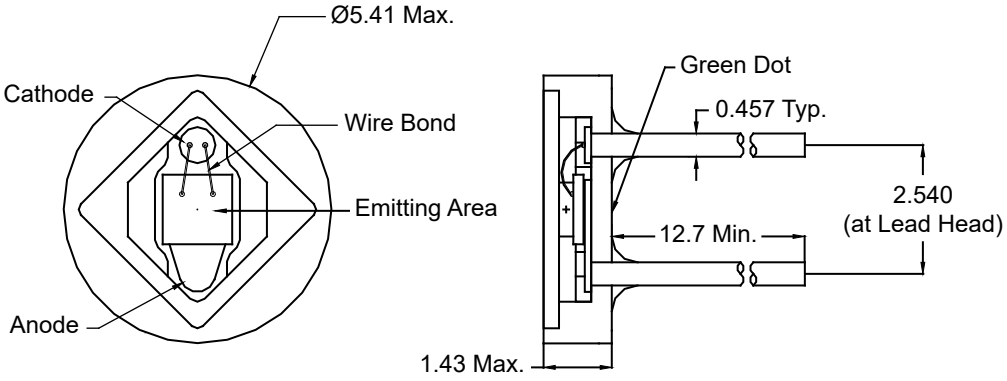
NDI does not offer a disc marker. The material selection, design choices, and mounting features will determine the characteristics of the disc marker, including whether you can sterilize it.

5.2 Active Markers

Active markers consist of an infrared light emitting diode mounted on a ceramic base. For active wired tools, the System Control Unit supplies power for the active marker; for active wireless tools, an external source, such as a battery or another device, supplies the power.

To ensure you correctly identify the connections, use the diode test function of a digital voltmeter. Figure 5-2 illustrates the marker components.

Figure 5-2 Standard Active Marker



5.3 Passive Tools

Passive tools are ideal for situations where electrical components on the tools are not possible or are unwanted. Passive tools tend to be faster and cheaper to develop than active wireless or active wired tools which require electrical compliance certification and additional safety measures.

Continuing on the information provided about passive tools in “Passive Tool Overview” on page 1, the following information must be considered.

Minimizing reflections

To minimize the reflections from the tool surface, consider the shape and finish of the tool. Reflections are particularly of concern for passive tools, which rely on infrared light reflected from the marker to the Position Sensor. Reflections from the tool surface can interfere with the Position Sensor’s ability to track and can reduce the accuracy of the reported transformations.

The shape and finish of the tool can affect the reflectivity of the tool.

Shape: A curved tool causes fewer reflections than a flat tool. A flat tool can act like a mirror and reflect back most of the light. A curved tool with a small radius will reflect light in different directions, so less light is reflected back to the Position Sensor.

Finish: Bead-blasting a tool before applying a finish reduces reflectivity. Bead-blasting roughens the surface so that less light is reflected toward the Position Sensor.

Note **The apparent reflectivity of a surface under visible light does not always indicate the reflectivity of the surface under infrared light. For example, matte black paint appears to absorb most light, but may still reflect infrared light. You can use the NDI ToolBox Capture utility to test the infrared reflectivity of the tool material.**

5.4 Active Wireless Tools

Active wireless tools don't require a physical connection to the tracking system. Synchronization between the active wireless tool and the Position Sensor is achieved via infrared signalling. However, active wireless tools require a power supply. Depending on the use case, power can be supplied from batteries or from an external power source.

Active wireless tools consist of an infrared light receiver, a system micro-controller, active markers, and a power supply. The NDI Active Wireless Development Kit design files and documentation can assist you with designing your own active wireless tools. For more information, see the *Polaris Active Wireless Tool Design Guide*.

5.5 Active Wired Tools

Active wired tools are ideal in situations where optically tracked tools must be small. Active markers have a smaller form factor than passive markers, and active markers tools can use standard geometries rather than unique geometries. Standard geometry tools tend to be smaller because they have fewer design constraints than unique geometry tools. Instead of relying on unique segment lengths and angles, a Polaris System connected to a System Control Unit can sequentially activate individual markers and use this temporal information to identify the tool. Active wired tools can also use faces or groups to avoid the minimum segment length requirement.

Active markers are also beneficial for reusable tools that are autoclavable.

Electrical Safety Considerations

It is beyond the scope of this document to describe all the electrical safety issues associated with tool design. Furthermore, the Position Sensor is only one component in the overall system — safety considerations must also be applied to the workstation, computer peripherals, isolation transformer, the suite of tools, and all other components of this system. It is the responsibility of the OEM to test and certify that the entire system complies with the necessary safety standards.

Polaris Systems with a System Control Unit are designed to be compatible with the medical safety standard IEC 60601. The tools are treated as F-Type Isolated (Floating) Applied Parts (Type BF). For this safety rating to apply, you must ensure that the tool design will maintain the isolation from electronic circuitry within the tool to objects that the tool may be expected to contact during its normal use.

To ensure that your tool design complies with all relevant electrical safety standards, remember the following:

- Work with an approval agency qualified in testing medical standards. Consult with the necessary safety approval agencies to obtain advice and guidance specific to a particular design.
- Consult and comply with the following standards:
  - **IEC 60601-1** Medical electrical equipment - Part 1: General requirements for safety
  - **IEC 60601-1-1** Medical electrical equipment - Part 1-1: General requirements for safety - Collateral standard: Safety requirements for medical electrical systems
  - **IEC 60601-1-2** Medical electrical equipment - Part 1-2: General requirements for safety - Collateral standard: Electromagnetic compatibility - Requirements and tests.
- Consult and comply with additional applicable national standards and amendments.
- Polaris Systems with a System Control Unit are designed for Type BF applied parts only.

Note **It is not straightforward to interpret the IEC 60601 standard as it applies to active tools, especially when these tools are connected to other electro-medical devices such as a surgical microscope or bipolar coagulating forceps. NDI recommends that you involve experts from the necessary safety approval agencies at the onset of the development project. This early involvement can help avoid an expensive re-design of the tool in order to comply with requirements of the medical standards.**

System Control Unit

To use active wired tools, you require a System Control Unit to provide the following functions:

- electrical isolation of the tools from the external power according to IEC 60601-1 for Type BF applied parts
- synchronization of the activation of active markers to the exposure period of the Position Sensor
- handling of auxiliary active wired tool functionality, such as user switch input and visible indicator LEDs

The Polaris Spectra uses a different System Control Unit than the one used for Polaris Vega and Polaris Lyra Systems. For more information about these, see the data sheet for the System Control Unit.

Switches and Visible LEDs

Active wired tools can be designed to include switches and visible light emitting diodes to enable communication and feedback between the system and user of the tool.

A switch on a tool can be used to trigger an action such as recording tracking data, or changing screens in the application software. Up to three switches can be incorporated into an active tool. When a switch is activated, the system sets a bit which can be read by the host computer. Switches are user-defined and must be incorporated into the tool wiring matrix shown in Figure 5-3.

A visible LED on a tool can be used to indicate information about the tool or system. For example, a visible LED could be placed on each face of a multi-face tool, to indicate which face is being tracked.

Up to four visible LEDs can be incorporated into an active wired tool. One visible LED, the tracking LED, is reserved for use by the system; the remaining LEDs are user-defined and controlled through your application.

A tool can have up to 20 markers, which includes visible LEDs. For example, if a tool uses four visible LEDs, there are 16 remaining positions available for active markers. A visible LED can be specified to be the tracking LED during characterization. A tracking LED is system-controlled and cannot be controlled by the API. The tracking LED is solid on when the tool is tracking and in the field of view of the Position Sensor (in or out of volume). The LED flashes when the system cannot detect enough markers to calculate a transformation or when the tool is out of the field of view.

Tool Wiring

When designing an active wired tool, you must choose the number and location of markers that are best suited to your application. Each active wired tool stores its tool definition file in an SROM device. You must consider the location of the SROM device and how the System Control Unit will recognize the tool. To reduce the possibility of signal loss, NDI recommends that you locate the SROM device inside the tool connector housing instead of in the tool body. For the System Control Unit to recognize the tool, the diode (D1 in the tool wiring matrix) must be in row 1 and must be connected to the field effect transistor. For more information, see Figure 5-3.

For tool designs that do not require any switch detection, another way to detect whether the tool is in the port is to populate marker position A (Column 1/Row 1) with an active marker. If the System Control Unit detects a not closed switch in row 1 (via D1), it tries to briefly activate marker A to determine if current is flowing. If this is the case, it interprets that as a tool being present. This check is done outside the marker firing period and does not affect tool tracking. If you use this method, you can omit both D1 and Q1 from the tool design.

Note **When choosing components, consider your sterilization requirements.**

Tool Wiring Matrix

The active markers and visible LEDs on an active tool are connected in a 4×5 matrix, as shown in Figure 5-3. Individual markers may be turned on by energizing the appropriate row and column. Multiple markers may be activated by supplying power to multiple rows or columns.

The Position Sensor decides which active markers to activate in a particular exposure period, and the System Control Unit is responsible for activating the markers within the exposure window. The activation of active markers is not continuous; instead, the System Control Unit cycles through each

of the four columns at high frequency, enabling the corresponding row in each column to activate the selected active markers. For optimal results, the same number of active markers must be fired in each column. If the number of active markers in each column is unbalanced, there will be a difference in brightness between the active markers on the tool, with the active markers in the column with the most markers appearing dimmer. This imbalance can lead to degradation in accuracy and repeatability of the tool transformation.

Example: Wiring a Four-Marker Tool

An optimal wiring would be to use A, G, M, and S in the tool wiring matrix. This way, the markers are distributed evenly across the columns of the tool wiring matrix and the markers are powered evenly.

A less-favourable way to wire a four-marker tool would be to use A, G, M, and B in the tool wiring matrix. In this case, markers A and B are in the same column of the tool wiring matrix and so would be significantly less bright than markers G and M and may not be detected.

Example: Wiring a Six-Marker Tool

An optimal wiring would be A, F, K, B, G, L. Not all columns in the wiring matrix are used, but the markers are distributed evenly across the columns that are used and all markers will have roughly the same brightness.

A sub-optimal way to wire the tool would be A, B, C, D, E, F. Since marker F is the only one in a column by itself, the A, B, C, D, and E markers would be weaker than F and may not be detected.

Example: Wiring a Five-Marker Tool

There is no way to distribute five markers evenly among four columns. Various tool designs that place two markers in some columns and one marker in others are equivalent.

Figure 5-3 Tool Wiring Matrix

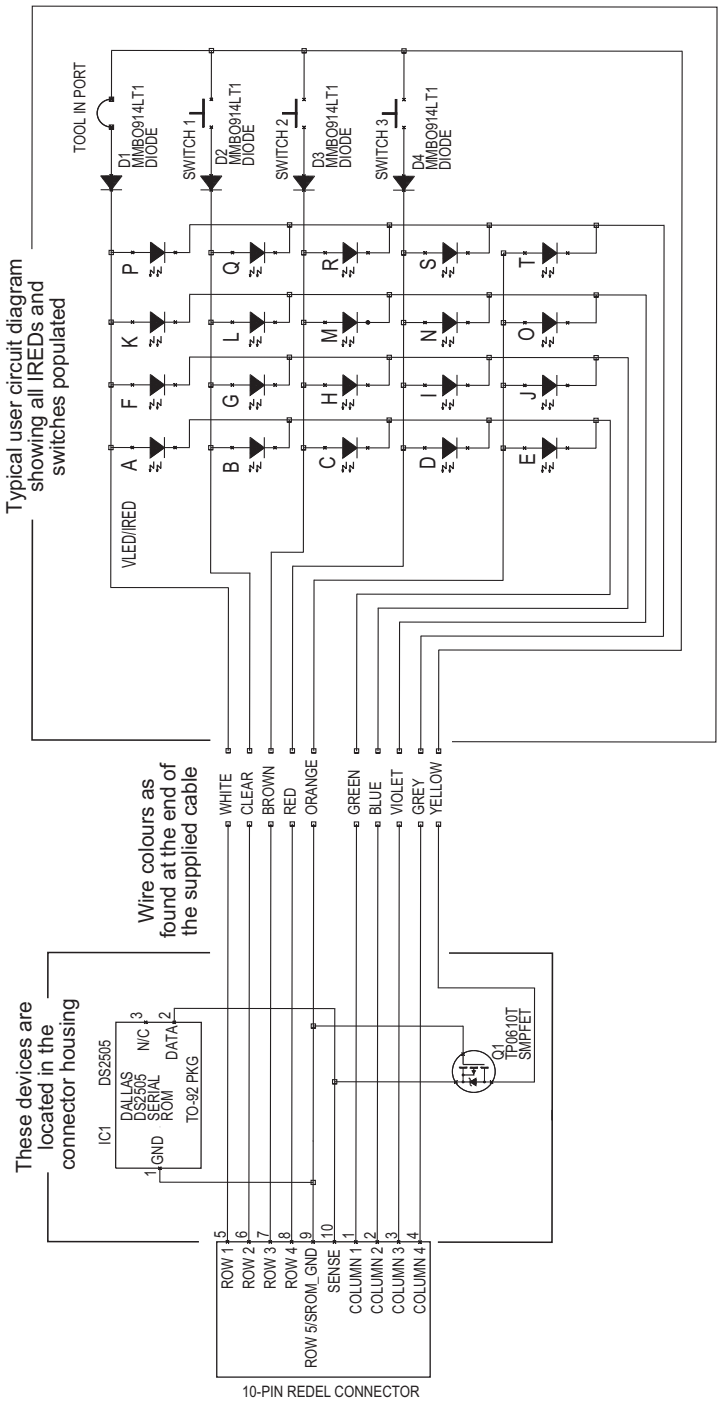
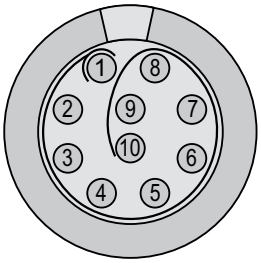


Figure 5-4 Tool Connector Pin Orientation -Viewed From Rear



Tool Cables

Tool cables and cable assemblies are available from NDI. The tool cable assembly wiring is shown in Figure 5-3 on page 27. The rear connection on NDI's tool cable assembly is not potted. If the rear connection is potted, the connector is suitable for autoclaving.

If you are considering using a different cable, there are many criteria that you must consider when selecting the cable. Many of these criteria are specific to the design and application and must be formulated by the designer. Two important criteria that must be considered in all applications are wire size and wire length.

The resistance of the wire used in the tool cable has an appreciable effect on the optical output power of the active marker. This resistance should be kept reasonably uniform across tool designs to ensure that the output power for the active markers will be approximately constant. The resistance of the wire from the tool connector to an individual active marker and back to the connector should be less than or equal to 1.50 Ω.

For example, suppose you wish to use a 26 gauge wire (65 strands of 44 gauge tinned copper) with a resistance of 44.1 Ω per 1000 feet. To calculate the maximum length of a cable that will have a total path resistance of 1.50 Ω, use:

$$2 \times \text{length} = \frac{1.5\Omega}{\left[\frac{44.1\Omega}{1000 \text{ feet}}\right]}$$

Thus, maximum cable length = 17.0 feet (5.18 m).

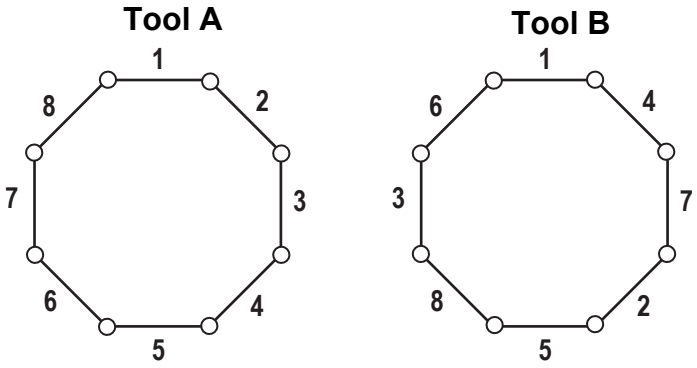
Marker Activation on Multi-Faced Tools

When the system is acquiring a multi-faced active tool for the first time or after it goes missing, it must activate each face of the tool in turn, starting with face 1, until it gets to a face that has enough markers in view to calculate a transformation. Certain marker activation sequences result in a faster average tool acquisition time.

Example: Marker Activation Sequence on an Eight-Face Tool

Figure 5-5 shows two possible activating orders for an eight-sided tool. On both tools, the markers are located at the corners of each face, so each marker is on two faces.

Figure 5-5 Multi-Faced Tool Activating Order



The Position Sensor can detect at least two faces at a time. On average, the system can find tool B faster than it can find tool A. For example, if the Position Sensor is facing both tools from side *5*, it would detect tool A when it activates the fourth side and tool B when it activates the second side.

6 Tool Tip Offset

When the system tracks a tool, it reports the transformations of the tool's local coordinate system relative to the Position Sensor's global coordinate system. The origin of the tool's local coordinate system is defined during tool characterization using Cygna-6D and is commonly defined as the average position of all the markers or the position of a single marker. However, it is often useful to track a different point of interest, such as the tip of a probe. You can define a tool's origin at the tip of a probe; however, if the tool is later bent, the origin is no longer located at the tip.

NDI recommends determining the tool tip offset (the vector between the tip of the tool and the origin of the tool) prior to each use. To determine the location of the tool tip, application software can apply the tool tip offset to the transformations.

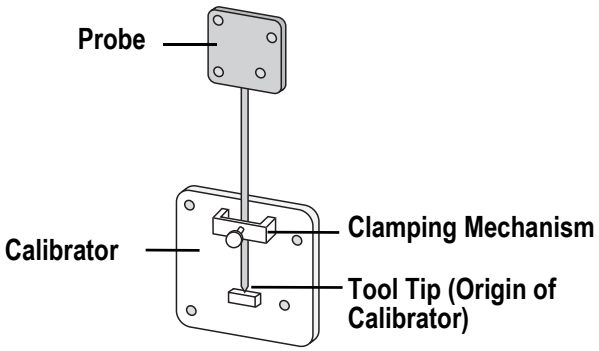
Determining the tool tip offset prior to each use ensures that the location of the tool tip is known as accurately as possible. You can determine the tool tip offset using a calibrator or by performing a pivoting procedure.

6.1 Calibrator

A calibrator is a rigid body that incorporates three or more markers and a clamping mechanism. The clamping mechanism allows you to clamp another tool (usually a probe) into place. Figure 6-1 shows an example of a calibrator.

To use a calibrator to determine the tool tip offset of a probe, clamp the probe in place on the calibrator. The point where the tool tip rests defines the origin of the calibrator. The system can then measure the positions of the probe's origin and the calibrator's origin. The application software compares these measurements to determine the tool tip offset of the probe.

Figure 6-1 Sample Calibrator



Calibrator Design Considerations

- The calibrator must follow the tool constraints outlined in "Markers and Tools" on page 21.
- The calibrator must have a groove in which the probe shaft will sit.
- The calibrator must have a clamp, to hold the probe in place in the groove. The probe must be held securely so that it will not move during the calibration data collection.

6.2 Pivoting

To determine the tool tip offset by pivoting, use NDI ToolBox or Cygna-6D. During the pivoting procedure, the system measures the positions of the markers while you pivot the tool. The software uses this data to determine the tool tip offset. If the tool has a ball tip, the tool tip offset represents the offset from the origin of the tool to the centre of the ball. If the tool has a sharp tip, the tool tip offset represents the offset from the origin of the tool to the tip of the tool.

To ensure that the pivoting procedure is accurate and repeatable, you must use a pivot block with an appropriately-sized divot.

Pivot Block Design Considerations

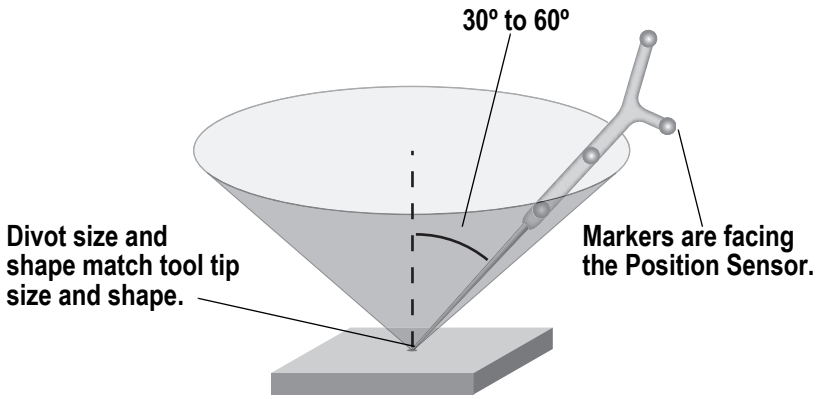
- The pivot block must be manufactured with substantial mass, to reduce the possibility of movement during use.
- It must be possible to securely clamp the pivot block.
- The pivot block must incorporate a divot, in which to pivot a tool. The size and shape of the divot must match the tool tip, to ensure that the tip does not move. For example, a probe with a 1 mm ball tip should have a hemispherical divot with a 1 mm diameter.
- A shallow divot must be provided for probe tools with sharp points.
- The pivot block may incorporate an appropriately sized convex arrangement (ball), to accommodate tools with concave tips.

Pivoting Technique

To ensure that the tip does not move while pivoting, the size and shape of the divot must match the tool tip.

1. Rest the tool tip in the divot in the pivot block.
2. Pivot the tool in a cone shape at an angle of 30º to 60º from the vertical, as shown in Figure 6-2.
    - Keep the tool tip stationary and ensure that there is a line of sight between the markers on the tool and the Position Sensor throughout the pivoting procedure.
    - Pivot the tool slowly for about 20 seconds.

Figure 6-2 Pivot Technique



7 Tool Accuracy

This chapter describes some of the factors that affect tool accuracy and how to improve the accuracy of the tool you are designing.

7.1 Marker Geometry

In general, the following may improve the accuracy of the returned tool transformations:

- **Increasing the number of markers:** Increasing the number of markers on a tool may improve the accuracy of the returned tool transformation, since it decreases the effect of each individual marker. If there are a large number of markers and one marker is measured inaccurately (for example, if it is partially occluded), its effect is minimized.
- **Increasing the 2D or 3D spread of the markers:** Increasing the distance between markers minimizes the effects of the error introduced by any one marker.
- **Decreasing the distance from the markers to the tool tip:** The shorter the distance from the markers to the tool tip, the smaller the extrapolation error.
- **Designing the tool so that the user can use it without occluding any markers:** Occluding markers can lead to the tool going missing if not enough markers are in view. It can also lead to less accurate transformations, due to fewer markers being used in the calculations or to a measured marker position being shifted from the actual marker position if the marker is only partially occluded. For more information on marker occlusion, see “Marker Occlusion Issues” on page 41.
- **Creating individual tool design files (.rom file):** Instead of relying on a generic tool design file, characterize the marker positions on each tool individually.

7.2 Tool Tracking Parameters

The tool definition file includes four tool tracking parameters, set when the tool is characterized. Figure 7-5 guides you through how the Polaris system uses the tool tracking parameters to determine which markers to use to calculate a tool transformation and when to return a transformation.

Note **Two physically identical tools using different parameters can have significantly different accuracy performance. It is important to carefully consider the effects of the tool tracking parameters.**

The tool tracking parameter values limit the magnitude of the RMS error returned with each transformation. This error is a measure of how well the system was able to fit the measured marker positions to the marker geometry specified in the tool definition file. Undesirably high RMS errors indicate that the tool tracking parameters were not set appropriately.

Maximum Marker Error

This parameter specifies the maximum allowable marker error for each marker on a tool. The marker error is the difference between the measured and expected location of a marker on a tool. The tool definition specifies the expected location of a marker on a tool.

How the system uses the maximum marker error: The system calculates an initial estimate for the tool transformation. It then performs a calculation to minimize the distance between the measured 3D position of each marker and the expected 3D locations as given in the tool definition file. A measure of this minimization is the RMS error reported with each transformation. The minimized 3D distance for each marker is the marker error. If the marker error for a particular marker is greater than the specified maximum marker error value, the system recalculates the tool transformation without the data from that marker.

How the maximum marker error improves tool accuracy: The maximum marker error parameter prevents the system from using potentially inaccurate data to calculate a tool transformation. For example, if a marker becomes partially occluded, the apparent centre of the marker shifts, which causes an error in the measured position of the marker. This error, in turn, increases the error in the calculated tool transformation. Setting a maximum marker error causes the system to stop using the marker data once the marker error has reached a certain level, reducing the possibility that the system will calculate inaccurate transformations.

Setting the maximum marker error: The value for the maximum marker error parameter must strike a balance between rejecting outlier marker measurements (for example, when the marker is partially occluded) and not rejecting valid marker measurements that will contribute to the accuracy of the tool transformations. The system can measure a marker throughout the characterized measurement volume with a particular accuracy; the maximum marker error parameter value must be set higher than the RMS accuracy of the system:

- **Vega (all models):** 0.12 mm RMS (pyramid volume) or 0.15 mm RMS (extended pyramid volume)
- **Lyra:** 0.20 mm RMS (pyramid volume) or 0.25 mm RMS (extended pyramid volume)
- **Spectra:** 0.25 mm RMS (pyramid volume) or 0.30 mm RMS (extended pyramid volume)
- **Vicra:** 0.25 mm RMS

Maximum Marker Angle

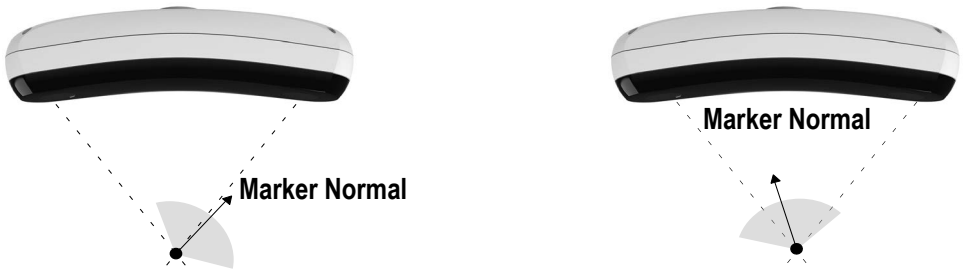
This parameter specifies the maximum allowable angle between a marker and each of the two sensors in the Position Sensor. Each marker in the tool definition file has an associated normal vector.

The system measures the angle between the marker normal and the vectors directed from the marker position to each sensor. If the angle between the marker normal and either sensor is greater than the specified maximum marker angle value, the system recalculates the transformation without the data from that marker.

In the diagram on the left side of Figure 7-1, the angle between the marker normal and the left sensor is greater than the maximum marker angle of 60 ° (highlighted region). Therefore, the system would not use this marker to calculate a transformation.

In the diagram on the right side of Figure 7-1, the angle between the marker normal and each sensor is less than the maximum marker angle of 60 °. Therefore, the system could use this marker to calculate a transformation.

Figure 7-1 Maximum Marker Angle



How the maximum marker angle improves tool accuracy: Markers (particularly active markers) do not behave as a perfect point source of light. As markers are viewed off-axis, the apparent centre of emission shifts. This is known as *isotropic error*, which increases as the off-axis viewing angle increases and can introduce significant errors in the tool transformations. Setting a maximum marker angle causes the system to stop using the marker data when the off-axis viewing angle reaches a certain level, reducing the possibility that the system calculates inaccurate transformations. The maximum marker angle can also be used to prevent the tool from tracking if there is a risk of partial marker occlusion.

Setting the maximum marker angle: When setting the maximum marker angle, consider whether markers might occlude each other. A tool may stop tracking due to partial marker occlusion before the markers exceed the maximum marker angle.

The maximum marker angle can be set in NDI Cygna-6D. The default maximum marker angle is 90º for an NDI Passive Sphere and 60º for an active marker or Radix Lens.

Checking the Marker Angles

The closer a marker is to the Position Sensor, the smaller the angle it can be turned and still be within the maximum marker angle. Use the following equation to check whether a marker is visible for a given position and orientation.

Let $\hat{n}$ be the marker normal, $\vec{m}$ be the marker position, and $\vec{s}$ be the sensor's position in the Position Sensor's coordinate system. The system will not use a marker to calculate the tool transformation if the marker angle (θ) for either sensor is greater than the maximum marker angle.

$$\theta = \arccos\left(\frac{\hat{n} \bullet (\vec{s} - \vec{m})}{\|\vec{s} - \vec{m}\|}\right)$$

For Polaris Vega and Spectra Systems, the positions of the sensors are approximately [0, -250, 0] and [0, 250, 0] mm.

For Polaris Lyra and Vicra Systems, the positions of the sensors are approximately [0, -100, 0] and [0, 100,0] mm.

Minimum Number of Markers

This parameter specifies the minimum number of markers that the system must use in the calculation of a tool transformation.

How the system uses the minimum number of markers: If the system cannot calculate a transformation using the minimum number of markers, it reports the tool as missing. The system may not be able to use the minimum number of markers, for example, if some of the markers are outside of the characterized measurement volume, are past the maximum marker angle, are occluded, or have exceeded the maximum marker error.

How the minimum number of markers improves tool accuracy: When the system calculates a transformation, it minimizes the overall error by fitting the measured marker positions, as closely as possible, to the marker geometry specified in the tool definition file. A measure of this minimization is the RMS error reported with each transformation.

Consider the case where one or more of the markers used in the transformation is inaccurate (for example, the marker is partially occluded). The system will calculate a transformation while minimizing the overall error. Although this fit may reduce the individual marker errors so that they lie within acceptable tolerances, it may also result in an error in the transformation.

Increasing the minimum number of markers reduces the effect of a single marker on the accuracy of a transformation. This is due to the fact that with more markers, the system will have more constraints when fitting the measured marker positions to a tool transformation if one of the markers is measured inaccurately. Thus, in general, more markers used to calculate a transformation results in a higher accuracy.

Setting the minimum number of markers causes the system to not report transformations when too few markers have been used in the calculation, reducing the overall error in reported transformations.

Setting the minimum number of markers: NDI recommends that the minimum number of markers be set to no less than four. If the minimum number of markers is set to three, the following situation can occur: suppose the system can detect three markers on a tool, and one measured marker position is shifted from the marker's actual position (due to a partial occlusion, for example). The three measured marker positions are still coplanar (since three markers are always coplanar), and so will then be fitted to an incorrect plane, which can significantly decrease the accuracy of the transformation.

If a tool is not designed to be a linear (5DOF) tool, the minimum number of markers must be greater than the greatest number of collinear markers on the tool.

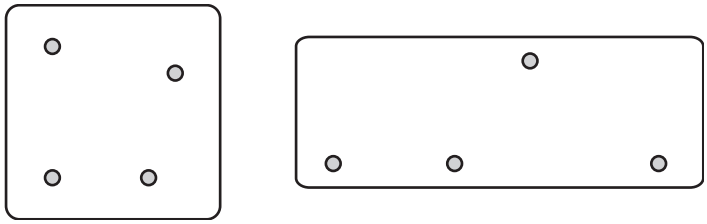
When setting the minimum number of markers parameter value, consider the performance of the tool for all possible combinations of that number of markers. "Estimating Tool Accuracy" on page 39 explains how to estimate tool accuracy of a probe tool. If specific combinations of markers result in a subsequent tool transformation accuracy that is unacceptable, consider the following courses of action:

- Redesign the marker geometry to eliminate this worst-case geometry.
- Increase the minimum number of markers parameter value.
- Use the minimum spread parameters to automatically reject particularly poor marker geometries. See "Minimum Spread" on page 36.

Setting the Minimum Number of Markers

Two four-marker planar tools are shown in Figure 7-2.

Figure 7-2 Setting the Minimum Number of Markers



Consider the tool on the left. A 6DOF transformation will be returned whether the minimum number of markers is set to either three or four. If the parameter is set to four, the transformations will be more accurate; however, if one marker is not used in the transformation (for example, if the marker is occluded) then the tool will be reported as missing.

Now consider the tool on the right. Three of the markers are collinear. To avoid a transformation being reported for the three colinear markers, you can:

- Set the minimum number of markers to four. If one marker is not used in the transformation (for example, if the marker is occluded), the system reports the tool as missing.
- Set the minimum number of markers to three and use the minimum spread parameters to prevent the system from returning a transformation that does not include the top marker. The spread parameters may be set such that one of the three collinear markers could be missing and the system can still return a transformation. See “Minimum Spread” on page 36.

Minimum Spread

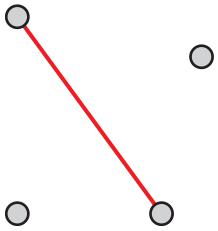
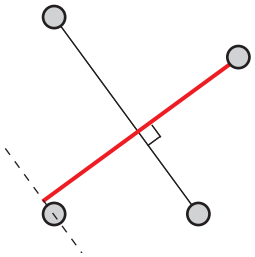
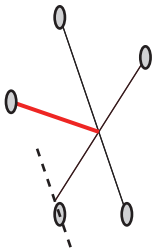
These parameters specify the distance, area, or volume that the markers used in the calculation of a tool transformation must cover. That is, they specify how much the markers need to be spread out.

How the system uses the minimum spread: After the system determines a final estimate for the tool transformation, it performs a marker spread check. The marker spread check calculates the minimum distance, area, or volume that would contain all the markers used to obtain this solution. It compares this result to the specified Minimum Spread 1, Minimum Spread 2, and Minimum Spread 3 parameters. If the space containing the markers used in the transformation is smaller than the space specified by the spread parameters, the system does not return the tool transformation. Table 7-1 describes how the system carries out the marker spread check for various values of the spread parameters.

Table 7-1 Marker Spread Check for Various Values of the Minimum Spread Parameters

Minimum Spread Parameter Values	Resulting Marker Spread Check	Example
Min. spread 1 = 0	The marker spread check is disabled.	

Table 7-1 Marker Spread Check for Various Values of the Minimum Spread Parameters

Minimum Spread Parameter Values	Resulting Marker Spread Check	Example
Min. spread 1 = non-zero Min. spread 2 = 0 Min. spread 3 = 0	The system performs a one-dimensional marker spread check. The system reports the transformation if the longest segment length (distance between two markers) on the tool is greater than the value of minimum spread 1.	
Min. spread 1 = non-zero Min. spread 2 = non-zero Min. spread 3 = 0	The system performs a two-dimensional marker spread check. The system reports the transformation if the following are met: • The largest segment length (distance between two markers) on the tool is greater than the value of minimum spread 1. • The largest vector perpendicular to the first segment has magnitude greater than the value of minimum spread 2.	
Min. spread 1 = non-zero Min. spread 2 = non-zero Min. spread 3 = non-zero	The system performs a three-dimensional marker spread check. The system reports the transformation if the following are met: • The largest segment length (distance between two markers) on the tool is greater than the value of minimum spread 1. • The largest vector perpendicular to the first segment has magnitude greater than the value of minimum spread 2. • The largest vector perpendicular to the first two has magnitude greater than the value of minimum spread 3.	

How the minimum spread improves tool accuracy: In general, the accuracy of tool transformations is improved by increasing the 2D or 3D distribution of markers on the tool. Setting the minimum spread parameters causes the system to only report transformations that include markers spread over a certain area or volume, improving the accuracy in reported transformations.

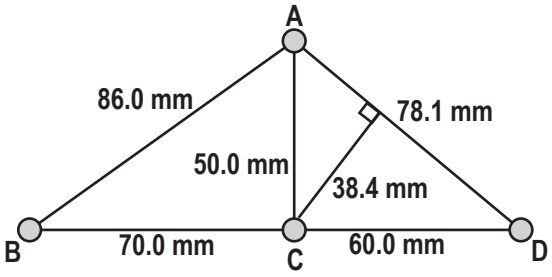
Setting the minimum spread parameters: Consider all possible marker geometries made up of the minimum number of markers, and set the spread parameters to reject poor marker geometries. Poor marker geometries might include, for example, collinear markers.

For a linear (5DOF) tool, spread parameters 2 and 3 must be set to zero, because the tool is one-dimensional. For a planar tool, spread parameter 3 must be set to zero, because the tool is two-dimensional.

Setting the Minimum Spread Parameters

The tool in Figure 7-3 shows a four-marker planar tool, with all the segment lengths labeled. Markers B, C, and D are collinear.

Figure 7-3 Four-Marker Planar Tool

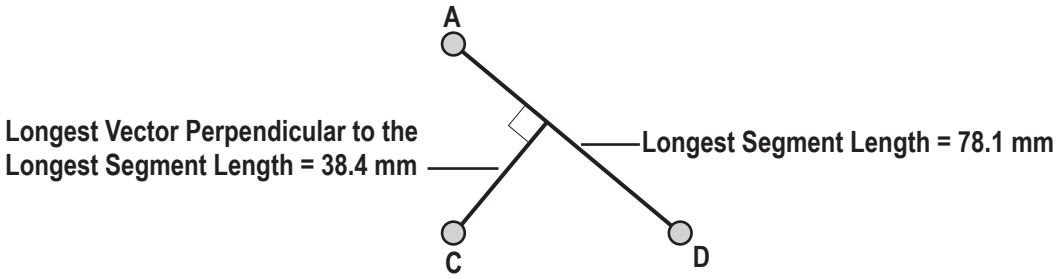


Consider the case where the minimum number of markers for this tool is set to three. The system will still return transformations even if one of the markers is not used to calculate the transformation (for example, if it becomes occluded). If, however, marker A is not used in the transformation, the tool becomes a linear (5DOF) tool. To avoid this possibility, the spread parameters can be set to make sure that transformations that do not include marker A are not reported.

Minimum spread 1 and minimum spread 2 must be set to non-zero values. This makes the system carry out a two-dimensional spread check, which will fail if marker A is not used.

To determine the values of marker spread 1 and marker spread 2, consider the smallest marker geometry that includes marker A but excludes one of the other markers. This is the case in which the system does not use marker B to calculate the tool transformation, as show in Figure 7-4.

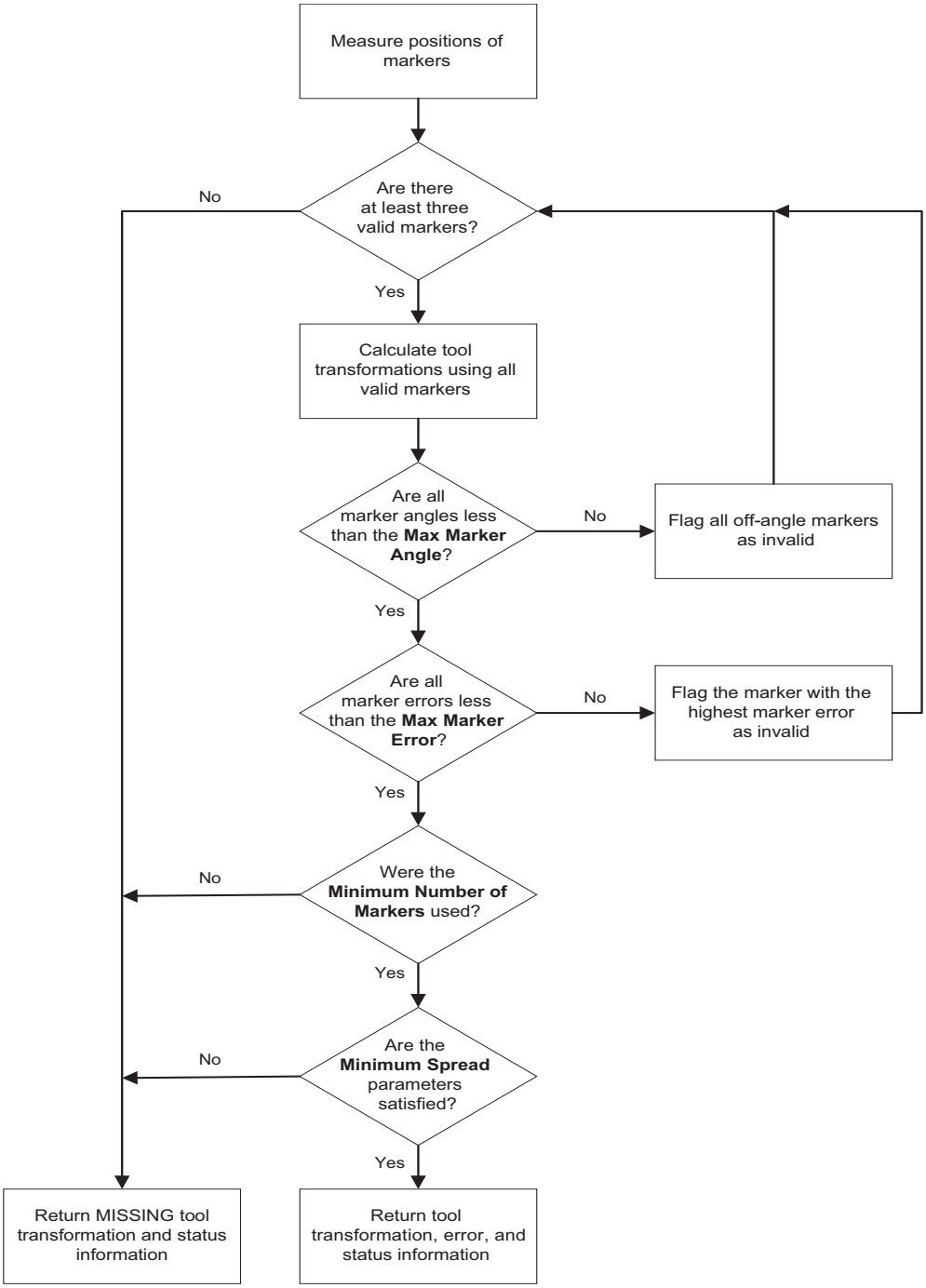
Figure 7-4 Minimum Spread Parameter



Thus, to allow for any one of the markers except marker A to be discarded, the spread parameters can be set to:

minimum spread 1 = 77
minimum spread 2 = 37.5
minimum spread 3 = 0

Figure 7-5 How the System Uses the Tool Tracking Parameters



7.3 Estimating Tool Accuracy

For information on the RMS accuracy of your system, see the user guide that accompanied your system. The actual accuracy of your system also depends on tool design.

Estimate Tool Accuracy

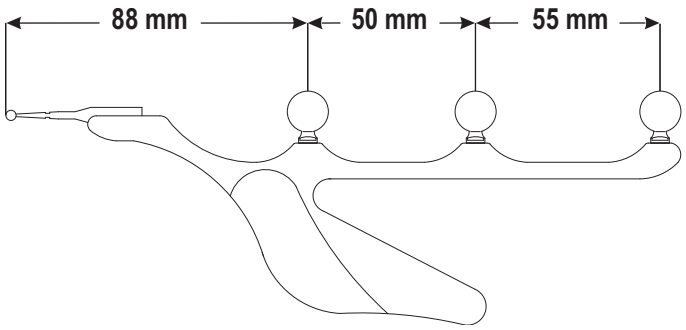
For a system that measures markers with an accuracy of *x* mm:

- If the tool has a point of interest (such as a probe tip) whose location will be calculated using application software, consider the maximum amount that point of interest could move while all the markers stay within *x* mm of their original positions.
- Be sure to estimate the accuracy of the tool for every combination of markers that fits the minimum number of markers and minimum spread parameter values that you intend to use in the tool definition file. If some combination of these markers leads to an undesirably high error, you should consider changing the parameter values or redesigning the tool.

Estimating the Accuracy of a Linear Probe

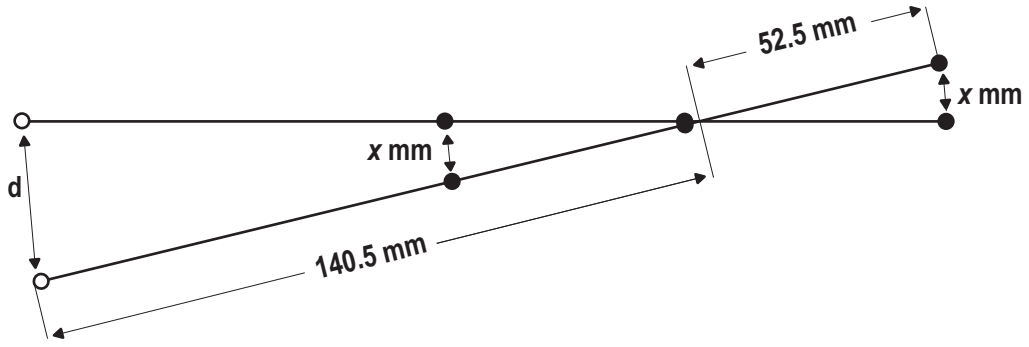
To estimate the accuracy of the linear probe tool in Figure 7-6, consider the maximum distance the probe tip could move while all the markers stay within *x* mm of their original positions, where *x* is the value you chose for the maximum marker error in the tool definition file.

Figure 7-6 Linear Probe



The worst-case scenario for the probe in Figure 7-6 is the case where the system reports the location of each of the end markers as being ±*x* mm from their actual position, in such a way that the reported position of the tool is rotated from its actual position about a point halfway between the end markers, as shown in Figure 7-7.

Figure 7-7 Estimating the Accuracy of a Linear Probe



The tool accuracy can be estimated by determining how far the reported tool tip location is from its actual location. This is distance **d** in Figure 7-7. The triangles on either side of the point of rotation in Figure 7-7 are proportional, so distance **d** can be determined using ratios: $\frac{\mathbf{d}}{140.5} = \frac{x}{52.5}$.

For example, consider a maximum marker error value of 0.6 mm. In this case, $x = 0.6$, so **d** = 1.6 mm. In the worst-case scenario, the reported probe tip location could differ from the actual probe tip location by as much as 1.6 mm.

A tool design which allows the most accurate tool tip offset has a large distance between markers along the tool's axis, and a short distance between the tool tip and the markers.

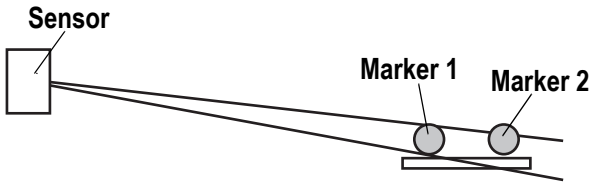
7.4 Marker Occlusion Issues

This section discusses the effects of markers occluding each other and offers some design suggestions to avoid marker occlusion.

What is Marker Occlusion?

Marker occlusion occurs when a marker is blocked (entirely or partially) from view of one or both of the sensors in the Position Sensor. This section deals with the specific situation where one marker occludes another marker. This type of marker occlusion occurs when two markers are approximately aligned with one of the sensors, as illustrated in Figure 7-8.

Figure 7-8 Marker Occlusion



When one marker partially occludes another, the markers appear to the sensor as though the infrared light sources are merged, as shown in Figure 7-9.

Figure 7-9 Sensor Image of Partially Occluded Marker



Marker occlusion is more likely to occur with NDI Passive Spheres; Radix Lens, active markers, and passive discs are less likely to occlude other markers due to their low profile form factor and reduced viewability.

Effects of Marker Occlusion

When one marker partially occludes another marker, either of the following results could occur:

- The system does not interpret the merged infrared light spots as being a valid marker. This results in one or both markers not being used to calculate the transformation.
- The system interprets the merged infrared light spots as being one of the markers. This results in the measured position of the marker being offset from the actual position of the marker. This offset may or may not exceed the maximum marker error parameter value.

The end result of the above situations can be one of the following:

- The system reports the tool as missing, because not enough markers were used to calculate the transformation.
- The system returns a transformation with an increased overall error, because one of the markers used in the transformation was partially occluded, and the measured position of this partially occluded marker was offset from its actual position.

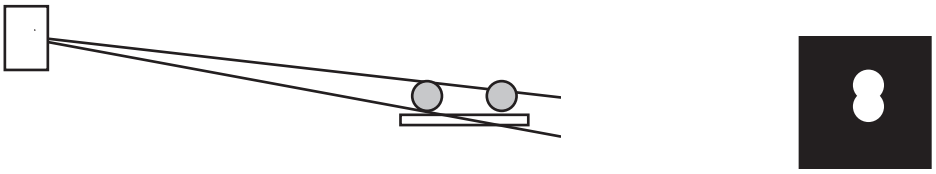
How to Avoid Marker Occlusion - Possible Solutions

A possible solution to avoid marker occlusion is to design the tool in such a way that it is not possible for the markers to occlude one another.

Example 1

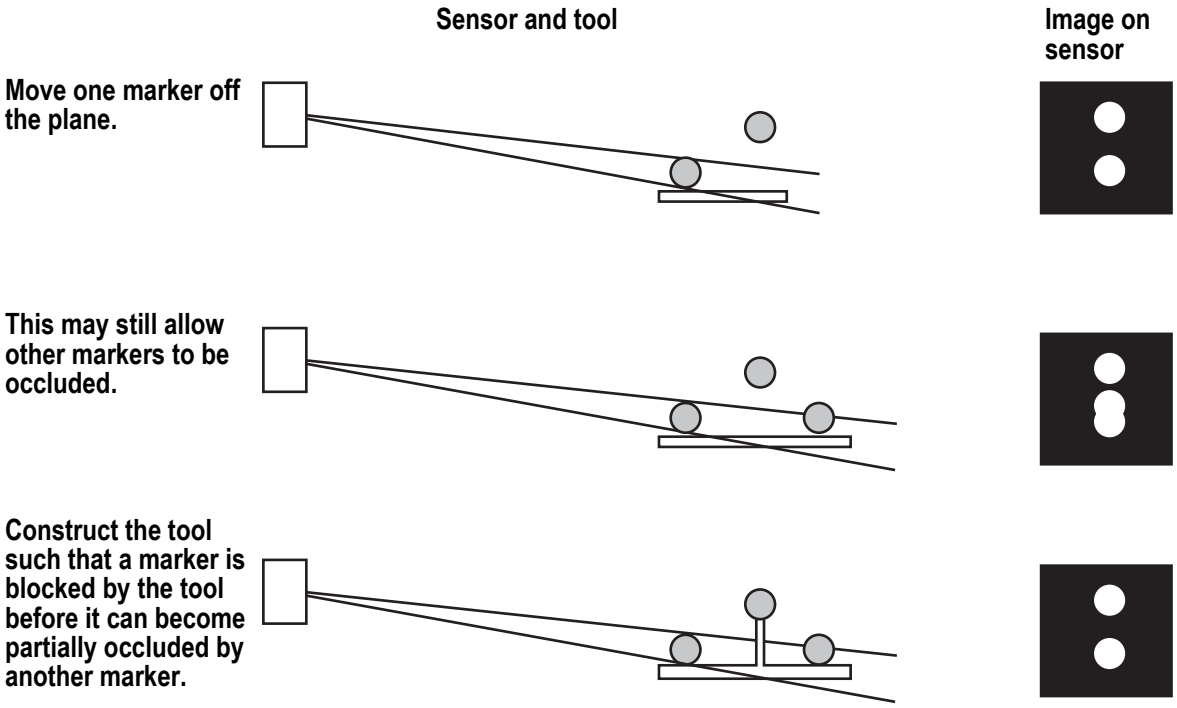
Consider the following planar tool. Figure 7-10 illustrates the side view and the sensor image view of a partial occlusion.

Figure 7-10 Marker Occlusion - Side View and Sensor Image



We can attempt to eliminate the occlusion by trying the following solutions:

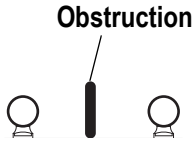
Figure 7-11 Eliminating Marker Occlusion



Example 2

You can build a tool such that there is an obstruction between every pair of markers, as illustrated in Figure 7-12. This obstruction will block the rear marker before it becomes occluded by the front marker. This eliminates the problem of one marker partially occluding another marker and resulting in a sensor image of merged infrared light sources.

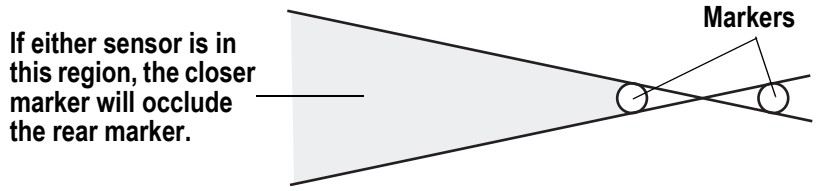
Figure 7-12 Obstruction Between Markers



The rest of this example explains how to determine the minimum size and location of such an obstruction.

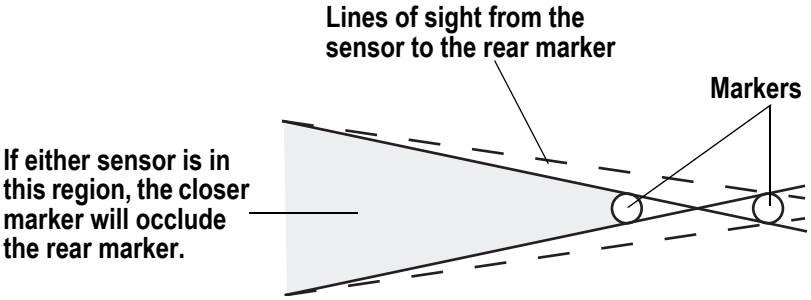
To determine the required size and location of the obstruction between a pair of markers, consider where the sensor can be positioned relative to the markers such that one marker partially occludes another. This is illustrated in Figure 7-13.

Figure 7-13 Sensor Location for Marker Occlusion



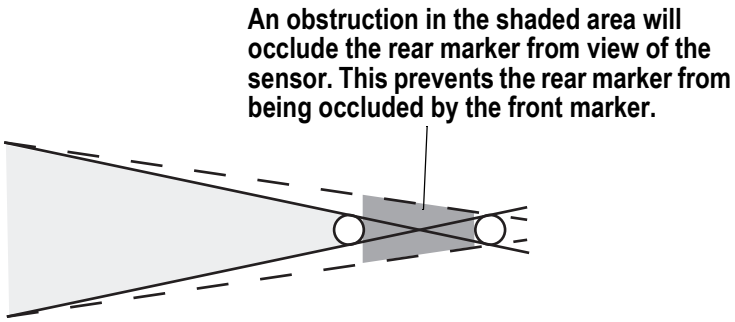
Now consider the lines of sight between the rear (partially occluded) marker and the sensor, when the sensor is in the zone highlighted in Figure 7-13. This is illustrated in Figure 7-14.

Figure 7-14 Lines of Sight from Sensor to Occluded Marker



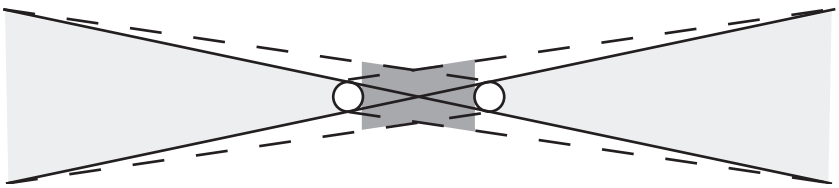
When those lines of sight are blocked, the front marker will never occlude the rear marker. We can place some obstruction between the two markers such that it blocks those lines of sight, as illustrated in Figure 7-15.

Figure 7-15 Prevent Marker Occlusion from One Side



The closer to the marker the rear obstruction is located, the smaller it needs to be. However, if the pair of markers may be viewed from the other side, the obstruction needs to be large enough to block the lines of sight from either side. The minimum size of the obstruction at various locations between the markers is illustrated in Figure 7-16.

Figure 7-16 Prevent Marker Occlusion from Any Side



The smallest obstruction is a disk placed exactly halfway between the two markers. The radius of the disk depends on the distance from the Position Sensor and the distance between the markers.

To stop the obstruction from partially occluding the tool (which can also lead to errors), set the maximum marker angle appropriately. An appropriate maximum marker angle stops the system from using the marker in the transformation before the marker can become partially occluded by the tool. This may mean setting the maximum marker angle parameter value to less than the default value. For more information, see “Maximum Marker Angle” on page 33.

8 Characterizing Tools

Characterization is the process of creating a tool definition file. A tool definition file is a file that describes a tool to the system. Without this information the system cannot accurately interpret the data it collects. You can characterize tools using Cygna-6D.

Note **After characterization, be sure to validate your tool to confirm it is operating as you expect and that it is returning correct transformations.**

8.1 Tool Definition Files

The characterization process creates a tool definition file.

- For passive and active wireless tools, you must load the file into the system before the system can track the tool.
- For active tools, the file is written to the SROM device within the tool at the end of the characterization process. You can also load a tool definition file for an active tool to the system to use instead of the file on the SROM device.

There are two ways to specify the location of the markers on a tool:

Collection: The system detects all the markers on the tool to accurately determine the marker positions.

Engineering Data: You specify the tool's marker positions, using the engineering specifications that are used to manufacture the tool. The system does not need to detect the tool when the tool is characterized using engineering data.

For active marker tools, mechanical tolerances and placement of markers prevent you from accurately knowing the location of the optical centre. Characterization using engineering data is not recommended.

For passive marker tools, the centre of the NDI Passive Sphere is the top of the mounting post and the centre of the Radix Lens is the centre of the top hemisphere. To use engineering data to characterize a tool, the accuracy placement tolerance for the mechanical centre point of the marker must not exceed 0.05 mm.

Note **For best accuracy for any tool type, use the collection method to characterize each tool individually.**

8.2 Characterization Best Practices

Consider the following best practices when you characterize tools. Poor characterization procedures may lead to a tool that produces inaccurate transformations, which may ultimately result in personal injury.

Before characterizing a tool

- Make sure the lenses on the Position Sensor are clean and not damaged.
- Make sure the markers on the tool are not damaged.

- Confirm the accuracy of the Position Sensor, for example using the Accuracy Assessment Kit (AAK), before characterizing tools.
- Make sure the environment is free of environmental IR that could interfere with the tool characterization process.

While characterizing a tool

- Orient the tool toward the Position Sensor to minimize off-angle markers.
- Take multiple data collections in different areas of the volume.

After characterizing a tool

- Make sure the environment is representative of the environment in which the tool will be used.
- Thoroughly test the tool and examine the tool RMS error. See “Testing Tools” on page 48.

9 Testing Tools

9.1 Simulating Aspects of the Design

Once you have designed a tool, you can create a scale 3D model of the tool and the Position Sensor using a mechanical CAD package to simulate aspects of the design.

One aspect of the design that may be effectively simulated is the requirement that a minimum number of markers with a sufficient geometric spread be seen by the Position Sensor throughout the specified range of tool positions and orientations. This can be done as follows:

1. In CAD software, place the tool at the desired position and orientation with respect to the Position Sensor.
2. Project a solid cone from each of the markers, pointing in the direction of the marker normal. The solid angle of the cone should be double the maximum marker angle.
3. If the cone corresponding to a particular marker envelops both sensors on the Position Sensor, then that marker will be visible at that position and orientation.

9.2 Testing the Final Design

After you have constructed a prototype for the tool, you can test the final design. The accuracy performance of the tool must be tested and documented through extensive bench testing. As a minimum, this analysis must consist of the following tests:

Test 1

1. Accurately move the tool to a measured set of representative positions and orientations. These measurements must be:
   - measured independently of the Position Sensor
   - traceable to an independent standard (e.g. National Institute of Standards and Testing or equivalent)

   These positions and orientations may be achieved with the use of precision translation and rotary stages or compound sine plates.
2. Compare the known positions and orientations of the tool with the positions and orientations measured by the Position Sensor.

Test 2

1. For each of the positions and orientations in test 1, test the effects of fully or partially occluding all possible combinations of markers. The degraded accuracy performance is tested by again comparing the measured and known positions and orientations.

Test 3

1. During testing, observe the RMS error that is returned with every transformation. The magnitude of the error is a function of the tool tracking parameter values as described in “Tool Tracking

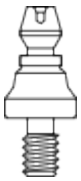
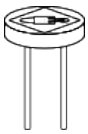
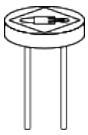
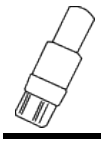
Parameters” on page 32. Undesirably high RMS errors indicate that the tool tracking parameters were not set appropriately.

There are many important protocols that might be selected to test each tool. In particular, the test must be able to discriminate between a tool that is performing marginally within and marginally outside specification. The variation in test results either repeated by the same tester or across different testers must be relatively small compared to the accuracy performance measurement itself.

Appendix A NDI-Supplied Parts

You can purchase the following parts from NDI:

Table 9-1 NDI-Supplied Tool Parts

Tool Part		Details
	NDI Passive Sphere	Single-use sterile NDI Passive Spheres Non-sterile passive spheres
	NDI Passive Sphere mounting post	See “NDI Passive Spheres” on page 21.
	Radix Lens	See “Radix Lenses” on page 21.
	Active marker	See “Active Markers” on page 22.
	Photodiode in ceramic package	See the *Polaris Active Wireless Tool Design Guide*.
	Tool cable	See “Tool Cables” on page 28.
	Tool cable assembly	See “Tool Cables” on page 28.
	SROM	See “Tool Wiring” on page 25.
	Tool connector	See “Tool Cables” on page 28.

Abbreviations & Acronyms

This guide uses the following abbreviations and acronyms:.

Abbreviation	Definition
5DOF	5 Degrees of Freedom
6DOF	6 Degrees of Freedom
CFR	Combined Firmware Revision
NDI	Northern Digital Inc.
RMS	Root Mean Square
SCU	System Control Unit
SROM	Serial Read-Only memory